# Infinite Series 1

Siddharth K

January 2026

## Problem 1

$$S = \sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)}$$

$$\frac{1}{(2n-1)(2n+1)} = \frac{1}{2}\left(\frac{1}{2n-1} - \frac{1}{2n+1}\right)$$

$$2S = \sum_{n=1}^{\infty} \frac{1}{2n-1} - \sum_{n=1}^{\infty} \frac{1}{2n+1} = \sum_{n=1}^{\infty} \frac{1}{2n-1} - \sum_{n=2}^{\infty} \frac{1}{2n-1} = 1$$

$$S = \frac{1}{2}$$

## Problem 2

$$S = \sum_{n=0}^{\infty} \frac{\cos n\theta}{2^n}$$

$$\cos n\theta = \frac{1}{2}(e^{in\theta} + e^{-in\theta}) = \frac{1}{2}((e^{i\theta})^n + (e^{-i\theta})^n)$$

$$2S = \sum_{n=0}^{\infty} \left(\frac{e^{i\theta}}{2}\right)^n + \sum_{n=0}^{\infty} \left(\frac{e^{-i\theta}}{2}\right)^n$$

$$2S = \frac{2}{2 - e^{i\theta}} + \frac{2}{2 - e^{-i\theta}}$$

$$S = \frac{1}{2 - e^{i\theta}} + \frac{1}{2 - e^{-i\theta}} = \frac{2 - e^{-i\theta} + 2 - e^{i\theta}}{(2 - e^{i\theta})(2 - e^{-i\theta})}$$

$$= \frac{4 - 2\cos\theta}{4 - 2(e^{i\theta} + e^{-i\theta}) + 1} = \frac{4 - 2\cos\theta}{5 - 4\cos\theta}.$$

# Problem 3

Let $k = n + 1$.

$$S = \sum_{k=1}^{\infty} \arctan \frac{2}{k^2} = \sum_{k=1}^{\infty} (\arctan(k+1) - \arctan(k-1))$$

$$S_N = \sum_{k=1}^{N} (\arctan(k+1) - \arctan(k-1)) = \arctan(N) + \arctan(N+1) - \arctan(0) - \arctan(1)$$

$$= \arctan(N) + \arctan(N+1) - \pi/4$$

$$S = \lim_{N \to \infty} S_N = 2 \cdot \frac{\pi}{2} - \frac{\pi}{4} = \frac{3\pi}{4}.$$

# Problem 4

$$S = \sum_{n=0}^{\infty} \frac{(n+1)^3}{n!}$$

$$\frac{(n+1)^3}{n!} = \frac{n^3}{n!} + \frac{3n^2}{n!} + \frac{3n}{n!} + \frac{1}{n!}$$

$$\sum_{n=0}^{\infty} \frac{1}{n!} = e$$

$$\sum_{n=0}^{\infty} \frac{n}{n!} = \sum_{n=1}^{\infty} \frac{1}{(n-1)!} = e$$

$$\sum_{n=0}^{\infty} \frac{n^2}{n!} = \sum_{n=2}^{\infty} \frac{n(n-1)}{n!} + \sum_{n=1}^{\infty} \frac{n}{n!} = \sum_{n=2}^{\infty} \frac{1}{(n-2)!} + e = 2e$$

$$\sum_{n=0}^{\infty} \frac{n^3}{n!} = \sum_{n=3}^{\infty} \frac{1}{(n-3)!} + 3 \sum_{n=2}^{\infty} \frac{1}{(n-2)!} + \sum_{n=1}^{\infty} \frac{1}{(n-1)!} = e + 3e + e = 5e$$

$$S = 5e + 3(2e) + 3e + e = 15e$$

# Problem 6

## Part (a)

$$F_n = \frac{1}{\sqrt{5}} \left( \left( \frac{1+\sqrt{5}}{2} \right)^n - \left( \frac{1-\sqrt{5}}{2} \right)^n \right) \to \frac{\varphi^n}{\sqrt{5}}$$

$$\lim_{n \to \infty} \sqrt[n]{|F_n x^n|} = \varphi |x|$$

By root test, if $|x| < \frac{1}{\varphi}$, the sum converges.

## Part (b)

$$S(x) = \sum_{n=0}^{\infty} F_n x^n, \quad xS(x) = \sum_{n=1}^{\infty} F_{n-1} x^n, \quad x^2 S(x) = \sum_{n=0}^{\infty} F_{n-2} x^n$$

Consider the coefficients of

$$S(x) - xS(x) - x^2 S(x).$$

For $n \geq 2$, the coefficients cancel to 0. $S(x)$ starts as $0 + 1 \cdot x + \cdots$, $xS(x)$ has no constant term and its $x^1$ coefficient is 0, and $x^2 S(x)$ has no $x^0$ or $x^1$ term. So

$$S(x) - xS(x) - x^2 S(x) = x,$$

and

$$S(x) = \frac{x}{1 - x - x^2}.$$

## Part (c)

From part (b),

$$S(1/10) = \frac{1/10}{1 - 1/10 - 1/100} = \frac{1}{8.9} = 0.1 \cdot \frac{1}{89},$$

where the sum is the Fibonacci sequence shifted by a decimal. Similarly,

$$S(1/100) = \frac{1}{9899}, \quad S(1/1000) = \frac{1}{998999}.$$